Research Article:

LASSO Regression – A Procedural Improvement

By

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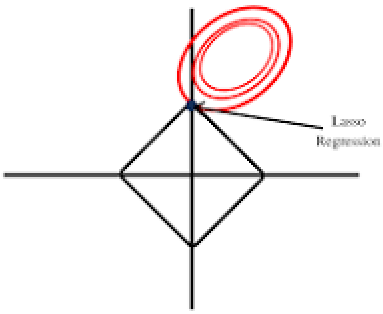
Abstract:

In high-dimensional data analysis, the LASSO regression is a potent statistical method for variable selection and regularization. Unlike conventional regression models, LASSO automatically shrinks irrelevant predictors to zero by minimizing the residual sum of squares while imposing a limit on the sum of the absolute values of the regression coefficients. LASSO is essential in domains like machine learning, bioinformatics, and finance because of its dual capacity to reduce dimensionality and improve prediction accuracy. The study emphasizes the value of LASSO regression in managing the complexity of contemporary datasets by connecting theory and practice. It improves predictive modeling's interpretability and dependability.

Keywords: Elastic Net, Hyperparameter Tuning, Machine Learning, Dimensionality Reduction, LASSO Regression, Variable Selection, Regularisation, High-Dimensional Data, Statistical Analysis, and Predictive Modelling.

1. Introduction:

Lasso regression, also known as Least Absolute Shrinkage and Selection Operator, is a statistical and machine learning method that selects and regularises variables to increase a model's accuracy and interpretability. In machine learning, it is frequently used to handle high-dimensional data. It lowers the chance of overfitting and is helpful when there is multicollinearity, or the correlation between several predictors. The primary function of LASSO (Least Absolute Shrinkage and Selection Operator), a statistical formula, is to select features and regularise data models. Geophysics was the first field to use it, and the phrase was first used in 1996 by Robert Tibshirani, a professor of statistics. Lasso was first developed for models of linear regression. This simple case reveals a substantial amount about the estimator.



Lasso regression works by:

1. Adding a penalty term to the residual sum of squares (RSS) is how Lasso regression operates.
2. The penalty term is multiplied by lambda (λ), the regularisation parameter.
3. Regulating the degree of regularisation used
4. Encouragement of sparsity in the model

As previously stated, Lasso regression is a regularisation method that uses a penalty to stop overfitting and improve statistical model correctness. The main purpose of Lasso regression, sometimes referred to as L1 regularisation, is to regularise linear regression models. Known as L1 regularisation, this kind of linear regression reduces or eliminates part of the model's coefficients by adding a penalty term to the cost function. *The penalty term is the absolute value of the coefficients multiplied by a scalar value called the regularization parameter.*

But, although lasso has a ℓ1 penalty rather than a ℓ2 penalty, it is quite similar to ridge regression. Similar to "Ridge Regression," "Lasso Regression" augments the linear regression objective function with a regularisation term. But in lasso regression, which employs L1 regularisation, the difference is in the loss function, which aims to minimise the sum of the absolute values of the coefficients multiplied by the penalty factor λ. The low magnitude of Lasso coefficients in comparison to ridge coefficients can be explained by the fact that they reach zero inside a specific range.[i]

2. Methodological Notes:

First, we see that a regression model using the L1 regularisation technique is known as lasso regression, while a model using the L2 regularisation technique is known as ridge regression. The least absolute shrinkage and selection operator (LASSO) is a regression analysis technique used in statistics and machine learning that combines variable selection with regularisation to improve the statistical model's predictability and interpretability.

2.1. It would be appropriate to note that sparsity in the model parameters is encouraged by L1 Regularisation (Lasso). It is possible for certain coefficients to become zero, thereby carrying out feature selection. In contrast, L2 Regularisation (Ridge) uniformly reduces the coefficients without necessarily bringing them to zero. Multicollinearity and model stability do benefit from it. It is a machine learning technique for supervised regularisation. We see that the type of data used is the primary distinction between supervised and unsupervised machine learning. Unlike unsupervised learning, supervised learning makes use of labelled training data. In other words, supervised learning models already know what the right output values need to be.

2.2. The fact that ChatGPT combines supervised and unsupervised learning is an extremely intriguing observation. For instance, we see that ChatGPT serves as an excellent benchmark for evaluating the relative advantages of supervised and unsupervised methods. The large language model (LLM) that powers ChatGPT, GPT-3.5, primarily employs unsupervised learning; however, more recent iterations of the model, such as Chat GPT 4.0, have also effectively integrated supervised learning.

3. The Formula for Lasso Regression:

The following minimisation problem results from using the LASSO estimator for linear regression, which is defined as the minimiser of a least squares objective function under the

L1 constraint for a data-dependent tuning parameter τ:

$$\text{Subject to } \sum^{k}{}_{j=1} |\beta_j| \leq \tau,\ \beta\hat{}\tau \text{ lasso} = \min \beta\{L(\beta)\}.$$

A generalisation of the lasso, the group lasso (Yuan & Lin, 2006) is primarily designed to enhance performance when predictors are grouped in some manner, such as when qualitative predictors are coded as dummy or one-hot variables (as is frequently done implicitly in ANOVA, for example).

4. Usefulness:

Common linear regression models are useful for forecasting results, which can assist businesses in determining whether to make particular investments or accept certain risks. Long-term business planning may be made easier as a result. Organisations can use this research, for instance, to figure out how many people can pass in front of a billboard. For instance, the healthcare industry provides real-world examples of lasso regression. For more precise diagnosis, Lasso Regression assists in extracting key characteristics from medical data. For instance, using patient information such as age, BMI, genetic markers, etc., to forecast the risk of diabetes.[1]

Lasso regression's primary goal is to promote sparse, basic models, i.e., models with fewer parameters. This kind of regression works well for models that exhibit significant levels of multicollinearity or when we wish to automate specific steps in the model selection process, such as parameter elimination and variable selection. Lasso regression reduces or eliminates the coefficients of insignificant predictors, simplifying the model and lowering its risk of overfitting. This enhances the model's readability and applicability to new data sets.[ii]

4.1. A note to overfitting/underfitting and lasso regression:

Overfitting in linear regression happens when the model is "too complex"—that is, when the number of parameters is significantly greater than the number of observations—or when the model specifications are in some other way, such as having redundant equations, etc. Typically, such a model will not perform well when applied to new data. It will, in other words, function well with training data but badly with test data. Regarding the potential of overfitting, it should be noted that the model may overweight particular features if there is multicollinearity among the explanatory variables. Additionally, the model becomes more sophisticated as a result, focussing on extraneous features and overfitting (fitting the dataset too closely). In this situation, lasso regression is helpful since it enforces a regularisation method that uses a penalty to stop overfitting and improve statistical model correctness. In this context, we observe that the VIF (variance inflation factor) detects the degree of multicollinearity. In general, multicollinearity may be present when the VIF is greater than 4 or the tolerance is less than 0.25, and more research is necessary. There is substantial multicollinearity that requires correction when VIF exceeds 10 or tolerance falls below 0.1.

4.2. Perhaps you're wondering why overfitting is seen negatively. Data scientists first train machine learning models on a known data set before using them to make predictions. The model then attempts to forecast results for fresh data sets using this information. An overfit model may not work well for all kinds of fresh data and may produce predictions that are not correct.

[1] https://shorturl.at/cerrq

4.3. It would not be inappropriate to discuss the factors that lead to underfitting. It happens when a model is overly basic, which can be caused by a model requiring more input characteristics, longer training time, or less regularisation. Similar to overfitting, underfitting causes training errors and subpar performance since the model is unable to identify the prevalent pattern in the data.

5. The difference between lasso regression and ridge regression:

Ridge regression keeps all of the model's features while decreasing the influence of less significant ones by making their coefficients smaller. By effectively choosing the most pertinent characteristics and enhancing model interpretability, Lasso regression can set some coefficients to zero. It should be mentioned that, in accordance with Hoerl, ridge regression—named for ridge analysis, where "ridge" denotes the path from the restricted maximum—is referred to as such in statistical literature.[2]

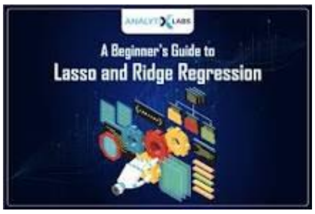
5.1. The absolute value of each coefficient is added to the loss function as a penalty term using L1 regularisation, also referred to as LASSO regression, as we have already discussed. The squared value of each coefficient is added to the loss as a penalty term via L2 regularisation, which is also referred to as Ridge regression. Therefore, how the penalty term is applied to the loss function is where the main distinction between lasso and ridge regression may be found. Therefore, it can be technically deduced that ridge regression operates by adding a term that penalises high coefficient values, whereas lasso regression is a type of regularisation that aims to reduce the magnitude of coefficients in order to include more pertinent variables in the model.

5.2. We observe that the ‘Lambda’ in lasso regression is the "Tuning Parameter" that regulates the bias-variance trade-off, and we use cross-validation to determine its optimal value. The purpose of this regularisation technique is to lessen overfitting. With one crucial exception—the Penalty Function is now equal to lambda*|slope|—it is comparable to ridge regression.

5.3. Before utilising Lasso and Ridge Regression, the variables must be normalised, or standardised. The size of the coefficients corresponding to each variable is constrained via Lasso regression. Nevertheless, the magnitude of each variable will determine this value.

[2] https://shorturl.at/nue3I

5.4. We also see that there is a distinction between regression and classification in this scenario. The primary distinction between classification and regression is that the former predicts discrete class labels, whereas the latter aids in the prediction of a continuous quantity. As a result, we occasionally discover some similarities between the two categories, particularly when discussing machine learning algorithms.[3]



6. Conclusion:

Applications in the healthcare industry and a few other extremely sensitive experimental engineering fields provide real-world examples of lasso regression. For more precise diagnosis, Lasso Regression assists in extracting key characteristics from medical data. For instance, estimating the risk of diabetes using patient information such as age, BMI, and genetic markers.

We might occasionally need to apply data augmentation with extreme caution. Augmenting data can be applied to assessment, training, or both. To a certain extent, using a lot of data improves deep learning performance. Similarly, expanding the kinds and quantity of small item samples in the dataset can also improve the detection performance of small objects.

Another term used in advanced regression is "endogeneity," which describes situations in which the relationship between an independent variable and a dependent variable cannot be interpreted carelessly due to the inclusion of omitted factors that result in biassed (i.e., contradictory) estimations (Antonakis et al., 2010). The Journal of Vocational Behaviour published this in 2020.

As briefly mentioned above, we draw the line here by assuming that Lasso Regression is a regularisation approach that, by adding a penalisation factor known as the alpha (or occasionally lamda) values, causes the data to shrink towards the centre point as the mean. The amount of

[3] https://shorturl.at/SwEUH

shrinkage (or constraint) that will be applied in the equation is indicated by the penalty term alpha (α).[iii]

7. References:

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[ii] https://tinyurl.com/2y7jwbrc
[iii] https://shorturl.at/vZj4u